DATA SCIENCE PORTFOLIO PROJECT

TELECOM CUSTOMER CHURN & RETENTION ANALYTICS

Customer behaviour • Churn risk • Retention opportunities

> **Headline result:** 26.54% of 7,043 customers churned; month-to-month and first-year customers showed the highest observed churn.

Project detail	Description
Dataset	IBM Telco Customer Churn (7,043 rows × 21 columns)
Tools	Microsoft Excel, MySQL, Python (Pandas, NumPy, Matplotlib, Seaborn), Power BI
Objective	Analyze customer churn, identify high-risk customer segments, and recommend data-driven retention strategies
Audience	Data science internship evaluators, recruiters, and technical interviewers
Report type	Descriptive analytics report (observed associations; no causal claims)
Document status	Draft for review before PDF export

Prepared as a portfolio project

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1. Executive Summary

This report analyses customer churn in the IBM Telco Customer Churn dataset. It measures overall churn and retention, compares churn across contract types, tenure groups, and selected service and payment segments, and translates the observed patterns into practical retention recommendations.

Overall KPI	Result
Total customers	7,043
Churned customers	1,869
Retained customers	5,174
Overall churn rate	26.54%
Retention rate	73.46%

Table 1. Overall customer KPIs.

Key observations:

- **Contract type.** Month-to-month customers have the highest observed churn rate (42.71%), compared with 11.27% for one-year and 2.83% for two-year contracts.
- **Tenure.** Customers in their first 12 months churn at 47.44%, falling to 9.51% for customers with 49–72 months of tenure.
- **Monthly charges.** Churned customers have a higher average MonthlyCharges (74.44) than retained customers (61.27).
- **Segments to investigate.** Fiber optic internet, electronic-check payment, and customers without online security or tech support show elevated churn and warrant further analysis.

> **Important:** All findings are descriptive associations. This analysis does not establish that contract type, tenure, charges, or any service feature independently causes churn.

The recommended priorities are early-stage onboarding and retention campaigns, carefully tested longer-term contract incentives, investigation of billing clarity, service quality and support experience, and ongoing monitoring of churn, retention, and intervention outcomes.

2. Project Objectives and Scope

2.1 Objectives

- Measure overall churn and retention using customer-level records.
- Compare churn rates across contract types and tenure groups.
- Compare average monthly charges between churned and retained customers.
- Examine additional segments such as internet service, payment method, online security, and tech support.
- Present results in an interactive single-page Power BI Executive Overview dashboard.
- Develop practical, testable retention recommendations.

2.2 Scope

The project covers descriptive analysis of one static dataset using four tools. It does not include predictive modelling, causal inference, customer-lifetime-value modelling, or controlled experiments.

Area	In scope	Out of scope
Analysis type	Descriptive KPIs and segment comparisons	Predictive or causal modelling
Data	IBM Telco Customer Churn dataset	External or time-series customer data
Outputs	Report, Excel, SQL, Python, Power BI	Deployed production system
Recommendations	Hypotheses and tests to run	Proven intervention results

Table 2. Project scope boundaries.

3. Dataset Overview

The source is the IBM Telco Customer Churn dataset, a customer-level dataset for a telecommunications provider. Each row represents one customer and includes account, service, billing, and churn-status information such as tenure, contract, internet service, payment method, MonthlyCharges, TotalCharges, and Churn.

Dataset attribute	Observed value
Rows (customers)	7,043
Columns	21
Unique customer IDs	7,043
Duplicate rows	0
Missing TotalCharges values	11

Table 3. Dataset shape and data-quality checks.

The target variable is Churn. A customer with Churn = Yes is counted as churned; Churn = No is counted as retained. Derived analysis fields used in this report include tenure groups (0–12, 13–24, 25–48, and 49–72 months).

4. Methodology and Tools

The analysis follows a layered workflow in which each tool plays a distinct role: spreadsheet exploration, database validation, programmatic analysis, and executive reporting.

Tool	Role in the project
Microsoft Excel	KPI summary, grouped comparisons, pivot-style analysis, and charts.
MySQL	Data validation and business queries using grouping, filtering, and segmentation.
Python	Pandas and NumPy for data inspection, quality checks, and churn calculations; Matplotlib and Seaborn for visualization.
Power BI	One-page interactive executive overview with KPI cards, charts, and slicers.

Table 4. Tools and responsibilities.

4.1 Python workflow

- Import libraries and load the cleaned CSV (telco_churn_cleaned.csv).
- Inspect shape, data types, missing values, duplicate rows, and unique customer IDs.
- Calculate the overall churn distribution.
- Analyze churn by contract type and tenure group.
- Compare MonthlyCharges between churned and retained customers.
- Visualize churn rates and document business insights.

4.2 Metric approach

Churn rate for any segment is calculated as churned customers in the segment divided by total customers in that segment. All percentages in this report are shown to two decimal places. KPI definitions are listed in Appendix A.

5. Data Quality and Preparation

Quality checks were run before any segment analysis. The dataset has no duplicate rows and every customer ID is unique, so each record can be treated as one customer.

Check	Result	Handling
Dataset shape	7,043 × 21	Confirmed
Unique customer IDs	7,043	No duplicate customers
Duplicate rows	0	No action needed
Missing TotalCharges	11	Retained and flagged for review; not silently deleted

Table 5. Data-quality summary.

5.1 Missing TotalCharges values

Eleven records have missing TotalCharges values. These records were not silently deleted. Any decision to impute, exclude, or analyze them separately should be documented explicitly and should preserve the original records. Because the churn, contract, tenure, and MonthlyCharges analyses in this report do not depend on TotalCharges, the headline KPIs retain all 7,043 customers.

5.2 Data-validation note: MySQL Churn field

> **Validation note:** During the MySQL import workflow, the Churn field showed a carriage-return normalization issue for the value “Yes”. SQL comparisons should use a normalized value (for example TRIM or a first-character check) until the field is cleaned and revalidated. This is an import-specific issue and should not be assumed to exist in the CSV loaded directly into Python or Power BI.

6. Key Findings

6.1 Overall churn

Of 7,043 customers, 1,869 churned and 5,174 were retained, giving an overall churn rate of 26.54% and a retention rate of 73.46%. Roughly one in four customers in the dataset churned.

6.2 Churn by contract type

Contract	Total customers	Churned customers	Churn rate
Month-to-month	3,875	1,655	42.71%
One year	1,473	166	11.27%
Two year	1,695	48	2.83%

Table 6. Churn by contract type.

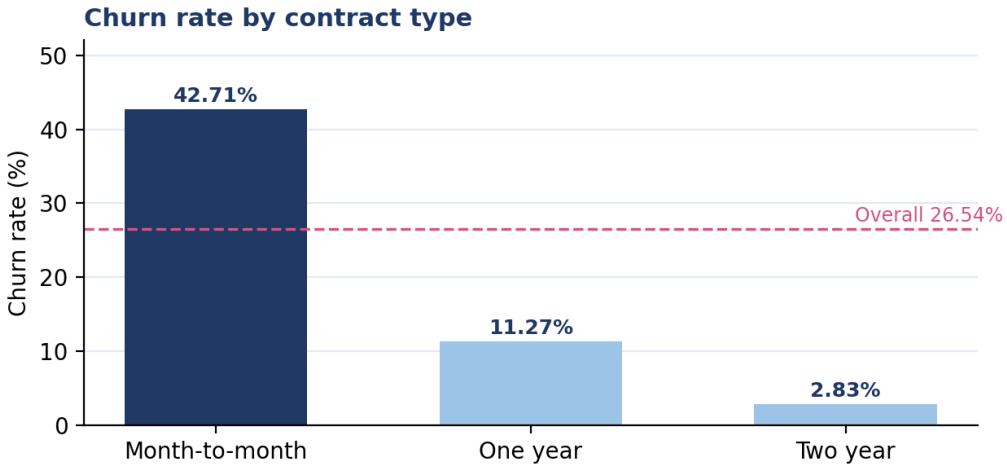


Figure 1. Contract-wise churn rate (dashed line = overall churn rate).

Month-to-month customers have the highest observed churn rate. They represent 55.02% of customers but 88.55% of all churned customers (derived from the table above). One-year and two-year contracts show much lower rates. Contract length is a useful segmentation variable for retention planning, but these figures alone do not show that contract type causes churn; customers who intend to stay may simply choose longer contracts.

6.3 Churn by tenure group

Tenure group	Total customers	Churned customers	Churn rate
0–12 months	2,186	1,037	47.44%
13–24 months	1,024	294	28.71%
25–48 months	1,594	325	20.39%
49–72 months	2,239	213	9.51%

Table 7. Churn by tenure group.

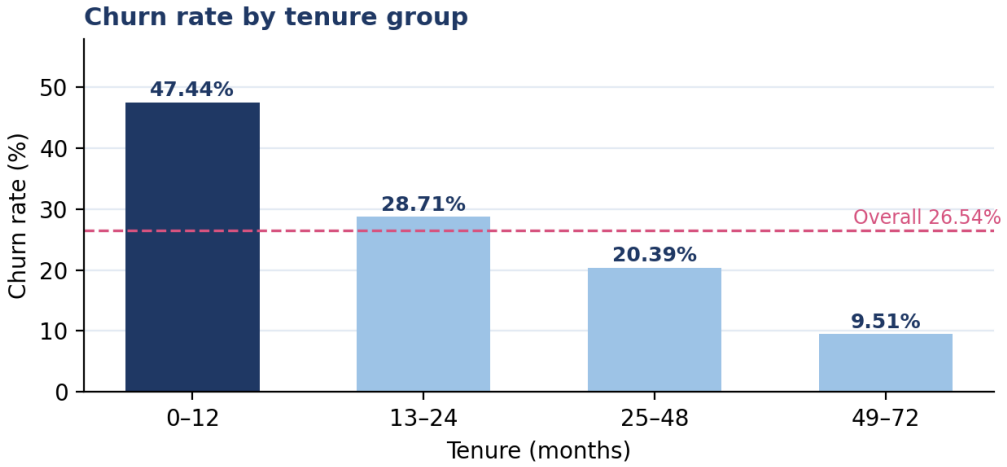


Figure 2. Tenure-wise churn rate (dashed line = overall churn rate).

Churn declines steadily as tenure increases. Customers in the first 12 months account for 31.04% of customers and 55.48% of churned customers (derived), making this the highest-priority segment in the observed data.

6.4 Monthly charges by churn status

Customer status	Average MonthlyCharges
Retained customers	61.27
Churned customers	74.44
All customers	64.76

Table 8. Average monthly charges by customer status.

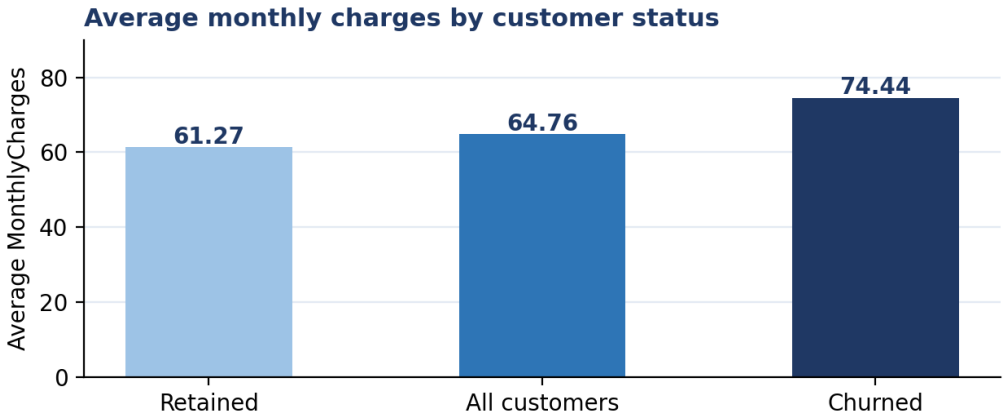


Figure 3. Average monthly charges for retained, all, and churned customers.

Churned customers pay 13.17 more per month on average than retained customers (about 21.50% higher, derived). This may reflect service mix, value perception, or support experience, but price alone should not be assumed to be the cause.

6.5 Additional segment observations

Dimension	Segment	Churn rate
Internet service	Fiber optic	41.89%
Internet service	DSL	18.96%
Internet service	No internet service	7.40%
Payment method	Electronic check	45.29%
Payment method	Automatic bank transfer	16.71%
Payment method	Automatic credit card	15.24%
Payment method	Mailed check	19.11%
Online security	Without online security	41.77%
Online security	With online security	14.61%
Tech support	Without tech support	41.64%
Tech support	With tech support	15.17%

Table 9. Churn by additional segments (highest-churn segment in each dimension highlighted).

Fiber optic customers churn at 22.93 percentage points more than DSL customers, and electronic-check users churn at 28.58 percentage points more than automatic bank-transfer users (derived).

These segments overlap and may reflect other customer characteristics, so they should be treated as hypotheses for deeper analysis before any targeting or pricing decision.

7. Business Interpretation

The findings point to where retention effort is most likely to matter. Each interpretation below is framed as an observed association with a suggested follow-up, not a proven cause.

Finding	What the data shows	Suggested interpretation
Contract risk	Month-to-month churn is 42.71% versus 2.83% for two-year contracts.	Month-to-month customers have the lowest commitment and are the largest at-risk pool.
Early-tenure risk	Churn is 47.44% in the first 12 months and 9.51% at 49–72 months.	The first year is a critical window for onboarding and service experience.
Charge level	Churned customers average 74.44 versus 61.27 for retained.	Value perception, service mix, or billing may deserve review; not proof of price sensitivity.
Service and payment	Elevated churn for fiber optic, electronic check, no online security, no tech support.	Candidate areas to examine for service quality, billing clarity, and support gaps.

Table 10. Business interpretation summary.

7.1 Association versus causation

This is a descriptive analysis of observational data. Customers are not randomly assigned to contract types, payment methods, or services, so differences between segments may be explained by other factors. For example, customers who already plan to stay may choose annual contracts, and segments such as fiber optic users may differ in price, usage, or location. Establishing causes would require controlled tests or more rigorous statistical modeling.

> **Reading guide:** Statements such as “higher churn among X” describe what was observed. They do not mean that X independently causes churn.

8. Recommendations

Recommendations are prioritized by the size of the observed churn gap. Each should be treated as a hypothesis to test and measure.

8.1 Early-stage onboarding and retention campaigns

Design an early-retention journey for customers in their first 12 months: onboarding check-ins, proactive issue resolution, service-use guidance, and satisfaction reviews. Define a baseline first-year churn rate and measure the change after launch.

8.2 Test longer-term contract incentives

Offer selected month-to-month customers clear, optional benefits for moving to longer contracts. Use a controlled test with a comparison group to confirm that the offer improves retention without eroding margin.

8.3 Investigate billing, service quality, and support

Review billing clarity, complaints, service quality, and support interactions for higher-charge customers and for the fiber optic, electronic-check, no-online-security, and no-tech-support segments. Avoid blanket discounts until the drivers of churn are better understood.

8.4 Monitor outcomes

Track churn rate, retention rate, first-year churn, contract mix, and monthly charges of churned customers over time, together with the outcomes of each intervention.

Priority	Action	Success measure
1	Early-tenure onboarding and retention campaign	First-year (0–12 month) churn rate
2	Contract-incentive test for month-to-month customers	Churn and margin versus control group
3	Billing, service, and support investigation	Findings documented; follow-up tests defined
4	Ongoing monitoring	Churn and retention rate trend; intervention results

Table 11. Recommendation roadmap.

9. Power BI Dashboard Design and Usage

The Power BI deliverable is designed as a single-page Executive Overview. It lets a business reader see headline churn performance at a glance and explore segments using slicers.

9.1 Dashboard components

Component	Purpose
KPI cards	Total Customers (7,043), Churned Customers (1,869), Retained Customers (5,174), Churn Rate (26.54%), Retention Rate (73.46%).
Contract-wise churn chart	Compares churn across month-to-month, one-year, and two-year contracts.
Tenure-wise churn chart	Shows how churn falls across the four tenure groups.
Monthly-charge analysis	Compares MonthlyCharges between churned and retained customers.
Customer-segment slicers	Lets the viewer filter the whole page by available customer segments.

Table 12. Executive Overview dashboard components.

9.2 Suggested layout

- Top row: five KPI cards, displayed as exact values (not abbreviated).
- Middle: contract-wise and tenure-wise churn charts side by side.
- Lower section: monthly-charge analysis.
- Side or top panel: segment slicers.

9.3 Usage guidance

- Start with slicers set to All; the cards should match the KPIs in Table 1.
- Select a segment to see how churn rate, counts, and charge patterns change.
- Read segment results as associations; small segments can produce unstable rates.

> **Validation status:** This report describes the dashboard design. The dashboard file itself has not been independently audited as part of this report. Before submission, confirm that card values match this report, that chart rates match Tables 6 and 7, and that slicers behave as expected.

10. Limitations

- The analysis is descriptive and does not establish causation.
- The dataset is a single static snapshot; findings may not generalize to other telecom companies, regions, or time periods.
- Segments overlap (for example, fiber optic and electronic-check customers), so single-factor comparisons can be confounded.
- The 11 missing TotalCharges values need documented handling if that field is used in further calculations.
- The MySQL Churn-field normalization issue means SQL comparisons need normalized values until the field is revalidated.

- No predictive model, statistical significance testing, or controlled experiment is reported.
- Completion and quality of individual project deliverables have not been independently audited in this report (see Appendix B).

11. Conclusion

Overall churn in the IBM Telco Customer Churn dataset is 26.54%, with retention at 73.46%. Churn is concentrated among month-to-month customers (42.71%) and customers in their first 12 months (47.44%), and churned customers have a higher average monthly charge (74.44 versus 61.27).

These results give a clear, evidence-based starting point for prioritizing early-tenure retention, testing contract incentives, and investigating billing, service, and support experience. Because the analysis is descriptive, the natural next step is to run controlled tests and, where appropriate, more rigorous modeling to learn which interventions actually reduce churn, and to monitor results over time.

Appendix A: KPI Definitions

KPI	Definition
Total Customers	Count of customer records in the analysis dataset.
Churned Customers	Count of records where Churn = Yes.
Retained Customers	Count of records where Churn = No.
Churn Rate	Churned Customers ÷ Total Customers (overall or within a segment).
Retention Rate	Retained Customers ÷ Total Customers.
Average MonthlyCharges	Mean of MonthlyCharges for the customer group being analyzed.
Tenure Group	Customer tenure in months, grouped as 0–12, 13–24, 25–48, and 49–72.

Table 13. KPI definitions.

Worked example: overall churn rate = 1,869 ÷ 7,043 = 26.54%; retention rate = 5,174 ÷ 7,043 = 73.46%.

Appendix B: Project Deliverables Checklist

Statuses below are deliberately conservative. Items are not marked complete unless verified; before submission, update this checklist against the actual project folder, verify all metrics, and export the reviewed report to PDF.

Deliverable	Expected filename	Status
Project report	01_Project_Report.docx / .pdf	Word draft; review, then export to PDF
Excel analysis	02_Excel_Analysis.xlsx	Not verified here; confirm
SQL queries	03_SQL_Queries.sql	Not verified here; confirm
Python EDA	04_Python_EDA.ipynb	Not verified here; confirm

Deliverable	Expected filename	Status
Power BI dashboard	05_PowerBI_Dashboard.pbix	Not independently audited; confirm
Presentation	06_Presentation.pptx	Pending / unverified
README	07_ReadMe.pdf	Pending / unverified

Table 14. Deliverables checklist (statuses to be confirmed by the author).